UNIT-III

PARTIAL DIFFERENTIAL EQUATIONS

1. $p\tan x + q\tan y = \tan z$

Sol) $p\tan x + q\tan y = \tan z$

Here $\Rightarrow P = \tan x,\ Q = \tan y,\ R = \tan z$

The auxilary eqn is $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

Consider, $\frac{dx}{\tan x} = \frac{dy}{\tan y}$

$$\int \frac{1}{\tan x}\,dx = \int \frac{1}{\tan y}\,dy$$

$$\int \cot x\,dx = \int \cot y\,dy$$

$$\log \sin x = \log \sin y + \log c$$

$$\log \sin x - \log \sin y = \log c$$

$$\frac{\sin x}{\sin y} = c$$

Let $u = \frac{\sin x}{\sin y}$.

Consider, $\frac{dy}{\tan y} = \frac{dz}{\tan z}$

$$\int \frac{1}{\tan y}\,dy = \int \frac{1}{\tan z}\,dz$$

$$\int \cot y\,dy = \int \cot z\,dz$$

$$\log \sin y = \log \sin z + \log c$$

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$$\frac{\sin y}{\sin z} = c$$

let $v = \frac{\sin y}{\sin z}$

The general solution is $\phi(u, v) = 0$

$$\Rightarrow \phi\left(\frac{\sin x}{\sin y}, \frac{\sin y}{\sin z}\right) = 0$$

solve $\frac{y^2 z}{x} p + xz q = y^2$.

$$\frac{y^2 z}{x} p + xz q = y^2$$

Here, $P = \frac{y^2 z}{x}$, $Q = xz$, $R = y^2$

The auxilary eqn is $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

$$\frac{dx}{\frac{y^2 z}{x}} = \frac{dy}{xz} = \frac{dz}{y^2}$$

consider, $\frac{dx}{\frac{y^2 z}{x}} = \frac{dz}{y^2}$

$$\int \frac{x\,dx}{y^2 z} = \int \frac{1}{y^2}\,dz$$

$$x\,dx = z\,dz$$

Integrate on both sides

$$\int x\,dx = \int z\,dz$$

$$\frac{x^2}{2} = \frac{z^2}{2} + c$$

$$x^2 - z^2 = 2c$$

let $u = x^2 - z^2$

Now, consider $\frac{dy}{xz} = \frac{dx}{\frac{yz}{x}}$

$$\frac{x}{y^2 z}\, dx = \frac{1}{xz}\, dy$$

$$x^2\, dx = y^2\, dy$$

Integrate on both sides,

$$\int x^2\, dx = \int y^2\, dy.$$

$$\frac{x^3}{3} = \frac{y^3}{3} + c$$

$$x^3 - y^3 = 3c$$

let $v = x^3 - y^3$

The general solution is $\phi(u, v) = 0$

$$\phi(x^2 - z^2,\ x^3 - y^3) = 0$$

3. $y^2 p - xyq = x(z - 2y)$

Sol) $y^2 p - xyq = x(z - 2y)$

Here $P = y^2$, $Q = -xy$, $R = x(z - 2y)$

The auxiliary eqn is $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

$$\frac{dx}{y^2} = \frac{dy}{-xy} = \frac{dz}{x(z - 2y)}$$

consider, $\frac{dx}{y^2} = \frac{dy}{-xy}$

$$\frac{1}{y}\, dx = -\frac{1}{x}\, dy$$

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$x\,dx = -y\,dy$

Integrate on both sides

$$\frac{x^2}{2} = -\frac{y^2}{2} + c$$

$$x^2 + y^2 = 2c$$

let $u = x^2 + y^2$

consider $\frac{dy}{-xy} = \frac{dz}{x(z-2y)}$

$$-\frac{1}{y}\,dy = \frac{1}{z-2y}\,dz$$

$$(z-2y)\,dy = -y\,dz$$

$$z\,dy - 2y\,dy = -y\,dz$$

$$z\,dy + y\,dz = 2y\,dy$$

$$d(yz) = 2y\,dy.$$

Integrate on both sides

$$\int d(yz) = 2\int y\,dy$$

$$yz = y^2 + c$$

$$yz - y^2 = c$$

let $v = yz - y^2$

The general solution is $\phi(u, v) = 0$

$$\phi(x^2 + y^2,\ yz - y^2) = 0$$

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4- Solve $x^2(y-z)p + y^2(z-x)q = z^2(x-y)$

Sol) $x^2(y-z)p + y^2(z-x)q = z^2(x-y)$

Here, $p = x^2(y-z)$, $q = y^2(z-x)$, $r = z^2(x-y)$

The auxilary eqn is $\frac{dx}{p} = \frac{dy}{q} = \frac{dz}{r}$

$$\frac{dx}{x^2(y-z)} = \frac{dy}{y^2(z-x)} = \frac{dz}{z^2(x-y)}$$

Choose multiplier $(1/x^2, 1/y^2, 1/z^2)$ then each ratio is

$$\Rightarrow \frac{\frac{1}{x^2}dx + \frac{1}{y^2}dy + \frac{1}{z^2}dz}{\frac{x^2}{x^2}(y-z) + \frac{y^2}{y^2}(z-x) + \frac{z^2}{z^2}(x-y)}$$

$$\Rightarrow \frac{\frac{1}{x^2}dx + \frac{1}{y^2}dy + \frac{1}{z^2}dz}{0}$$

$$\Rightarrow \frac{1}{x^2}dx + \frac{1}{y^2}dy + \frac{1}{z^2}dz$$

$$\int \frac{1}{x^2}dx + \int \frac{1}{y^2}dy + \int \frac{1}{z^2}dz = \log c$$

$$-\frac{1}{x} - \frac{1}{y} - \frac{1}{z} = c$$

$$\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = -c$$

let $u = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$

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choose multiplier $(1/x, 1/y, 1/z)$. then the ratio is

$$\Rightarrow \frac{\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz}{x(y-z)+y(z-x)+z(x-y)}$$

$$\Rightarrow \frac{\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz}{xy-xz+yz-xy+xz-yz}$$

$$\Rightarrow \frac{\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz}{0}$$

$$\Rightarrow \frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz$$

$$\Rightarrow \int \frac{1}{x}dx + \int \frac{1}{y}dy + \int \frac{1}{z}dz = \log c$$

$$\Rightarrow \log x + \log y + \log z = \log c$$

$$\Rightarrow xyz = c$$

let $v = xyz$

The general solution is $\phi(u, v) = 0$

$$\phi\left(\frac{1}{x}+\frac{1}{y}+\frac{1}{z}, xyz\right) = 0$$

5. solve $px^2 - qy^2 = z(x-y)$

Sol) $px^2 - qy^2 = z(x-y)$

Here, $p = x^2$, $q = -y^2$, $r = z(x-y)$

The auxilary eqn is $\frac{dx}{p} = \frac{dy}{q} = \frac{dz}{r}$

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choose multiplier $(1/x, 1/y, -1/z)$ then the ratio is

$$\frac{\frac{1}{x}dx + \frac{1}{y}dy - \frac{1}{z}dz}{x-y-x+y}$$

$$\Rightarrow \frac{\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz}{0}$$

$$\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{z}dz$$

$$\int \frac{1}{x}dx + \int \frac{1}{y}dy - \int \frac{1}{z}dz = \log c$$

$$\log x + \log y - \log z = \log c$$

$$\frac{xy}{z} = c$$

let $u = \frac{xy}{z}$

consider, $\frac{dx}{x^2} = -\frac{dy}{y^2}$

$$\int \frac{1}{x^2}dx = -\int \frac{1}{y^2}dy$$

$$-\frac{1}{x} = \frac{1}{y} + c$$

$$\frac{1}{y} + \frac{1}{x} = -c$$

let $v = \frac{1}{x} + \frac{1}{y}$

The general solution $\phi(u, v) = 0$

$$\phi\left(\frac{xy}{z}, \frac{1}{x} + \frac{1}{y}\right) = 0$$

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6. $(z^2 - 2yz - y^2)p + (xy + xz)q = (xy - xz)$

Sol) $(z^2 - 2yz - y^2)p + (xy + xz)q = (xy - xz)$

Here, $P = z^2 - 2yz - y^2$, $Q = xy + xz$, $R = xy - xz$

The auxilary eqn is $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

$$\Rightarrow \frac{dx}{z^2 - 2yz - y^2} = \frac{dy}{xy + xz} = \frac{dz}{xy - xz}$$

consider, $\frac{dy}{xy + xz} = \frac{dz}{xy - xz}$

$$\frac{dy}{x(y + z)} = \frac{dz}{x(y - z)}$$

$$(y - z)\,dy = (y + z)\,dz$$

$$y\,dy - z\,dy = y\,dz + z\,dz$$

$$y\,dy - z\,dz = y\,dz + z\,dy$$

$$d(yz) = y\,dy - z\,dz$$

$$\int d(yz) = \int y\,dy - \int z\,dz$$

$$yz = \frac{y^2}{2} - \frac{z^2}{2} + c$$

$$2yz + z^2 - y^2 = 2c$$

let $u = 2yz + z^2 - y^2$

Choose multiplier (x, y, z) then the ratio is

$$\frac{x\,dx + y\,dy + z\,dz}{xz^2 - 2xyz - xy^2 + xy^2 + yxz^2 + x^2yz - xz^2}$$

$$\Rightarrow \frac{x\,dx + y\,dy + z\,dz}{0}$$

$$\Rightarrow \int x\,dx + \int y\,dy + \int z\,dz = c$$

$$\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = c$$

$$x^2 + y^2 + z^2 = 2c$$

Let $\quad v = x^2 + y^2 + z^2$

The general solution is $\phi(u, v) = 0$

$$\phi(2yz - y^2 + z^2,\ x^2 + y^2 + z^2)$$

7) $(mz - ny)p + (nx - lz)q = (ly - mx)$

Sol) $(mz - ny)p + (nx - lz)q = (xy - xz)$

Here $\quad p = mz - ny,\ q = nx - lz,\ r = xy - xz$

The auxiliary eqn is $\frac{dx}{p} = \frac{dy}{q} = \frac{dz}{r}$

$$\Rightarrow \frac{dx}{mz - ny} = \frac{dy}{nx - lz} = \frac{dz}{xy - mxz}$$

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choose multipliers (x, y, z) then the ratio is

$$\frac{x\,dx + y\,dy + z\,dz}{xlz - nxy + nxy - lxy + lyz - xz^2 nl}$$

$$\Rightarrow \frac{x\,dx + y\,dy + z\,dz}{0}$$

$$\Rightarrow \int x\,dx + \int y\,dy + \int z\,dz = c$$

$$\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = c$$

$$x^2 + y^2 + z^2 = 2c$$

let $u = x^2 + y^2 + z^2$

choose multipliers (l, m, n) then the ratio is

$$\frac{l\,dx + m\,dy + n\,dz}{lmz - lny + mnx - mlz + lmy - lmx}$$

$$\Rightarrow \frac{l\,dx + m\,dy + n\,dz}{0}$$

$$\Rightarrow \int l\,dx + \int m\,dy + \int n\,dz = c$$

$$lx + my + nz = c$$

let $v = lx + my + nz$

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The general solution is $\phi(u,v) = 0$

$\phi(x^2+y^2+z^2, lx+my+nz) = 0$

8. $(x^2-y^2-z^2)p + 2xyq = 2xz$

Sol) $(x^2-y^2-z^2)p + 2xyq = 2xz$

Here $P = x^2-y^2-z^2$, $Q = 2xy$, $R = 2xz$

The auxiliary eqn is $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

$$\Rightarrow \frac{dx}{x^2-y^2-z^2} = \frac{dy}{2xy} = \frac{dz}{2xz} \quad -(1)$$

consider, $\frac{dy}{2xy} = \frac{dz}{2xz}$

$$\frac{1}{2}\frac{1}{y}dy = \frac{1}{2}\frac{1}{z}dz$$

$$\frac{1}{2}\int\frac{1}{y}dy = \frac{1}{2}\int\frac{1}{z}dz$$

$$\log y = \log z + \log c$$

$$\log\left(\frac{y}{z}\right) = \log c$$

$$\frac{y}{z} = c$$

Let $u = y/z$

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choose multipliers (x, y, z) then the ratio is

$$\frac{x\,dx + y\,dy + z\,dz}{x^3 - xy^2 - xz^2 + 2x^2y^2 + 2xz^2}$$

$$\Rightarrow \frac{x\,dx + y\,dy + z\,dz}{x(x^2 + y^2 + z^2)}$$

compare this with eq (1)

$$\frac{dz}{2xz} = \frac{x\,dx + y\,dy + z\,dz}{x(x^2 + y^2 + z^2)}$$

$$\frac{dz}{z} = \frac{2x\,dx + 2y\,dy + 2z\,dz}{(x^2 + y^2 + z^2)}$$

$$\frac{dz}{z} = \frac{d(x^2) + d(y^2) + d(z^2)}{x^2 + y^2 + z^2}$$

$$\frac{dz}{z} = \frac{d(x^2 + y^2 + z^2)}{x^2 + y^2 + z^2}$$

Integrating on both sides

$$\int \frac{1}{z}\,dz = \int \frac{1}{x^2 + y^2 + z^2}\, d(x^2 + y^2 + z^2)$$

$$\log z = \log(x^2 + y^2 + z^2) + \log c$$

$$\frac{z}{x^2 + y^2 + z^2} = c$$

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let $u = \frac{z}{x^2+y^2+z^2}$

The general solution is $\phi(u,v) = 0$

$$\phi\left(y/z,\ z/x^2+y^2+z^2\right) = 0$$

9. $4r + 12s + 9t = 0$

Sol. Given,

$4r + 12s + 9t = 0$

Here, $r = \frac{d^2z}{dx^2}$, $s = \frac{d^2z}{dxdy}$, $t = \frac{d^2z}{dy^2}$

$\therefore 4D^2 + 12DD' + 9D'^2 = 0$

The $f(D, D')z = 0$

The auxiliary eqn is $f(m, 1) = 0$

$$4m^2 + 12m(1) + 9(1)^2 = 0$$

$$4m^2 + 12m + 9 = 0$$

$$4m^2 + 6m + 6m + 9 = 0$$

$$2m(2m+3) + 3(2m+3) = 0$$

$$m = \frac{-3}{2}, -\frac{3}{2}$$

C.F $= \phi_1(y - \frac{3}{2}x) + \phi_2 x(y - \frac{3}{2}x)$

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10. $\frac{\partial^3 z}{\partial x^3} - 3\frac{\partial^3 z}{\partial x^2 \partial y} + 4\frac{\partial^3 z}{\partial y^3} = e^{x+2y}$.

Sol) $\frac{\partial^3 z}{\partial x^3} - 3\frac{\partial^3 z}{\partial x^2 \partial y} + 4\frac{\partial^3 z}{\partial y^3} = e^{x+2y}$.

$$D^3 - 3D^2D' + 4D'^3 = e^{x+2y}$$

$\Rightarrow f(D, D')z = f(x, y)$

The auxiliary eqn is $f(m, 1) = 0$

$$m^3 - 3m^2(1) + 4(1)^3 = 0$$

$$m^3 - 3m^2 + 4 = 0$$

	1	-3	0	4
-1		-1	4	-4
	1	-4	4	0

$$(m-2)(m^2 - m - 2) = 0$$

$$(m-2)(m^2 - 2m + m - 2) = 0$$

$$(m-2)(m(m-2) + 1(m-2)) = 0$$

$$m = -1, 2, 2$$

C.F = $\phi_1(y-x) + \phi_2(y+2x) + x\phi_3(y+2x)$

P.I = $\frac{1}{f(D, D')} F(x, y)$

$= \frac{1}{D^3 - 3D^2D' + 4D'^3} e^{x+2y}$

$= \frac{1}{(1)^3 - 3(1)^2(2) + 4(2)^3} e^{x+2y}$.

$$= \frac{1}{1-6+32} e^{x+2y}$$

$$= \frac{1}{27} e^{x+2y}$$

The general solution $z = CF + PI$

$$z = \phi_1(y-x) + \phi_2(y+2x) + \phi_3 x(y+2x) + \frac{1}{27} e^{x+2y}$$

11. $(D^2 - DD')z = \sin x \cos 2y$.

Sol) $(D^2 - DD')z = \sin x \cos 2y$.

$$f(D, D')z = F(x, y)$$

The auxiliary eqn is $f(m, 1) = 0$

$$m^2 - m(1) = 0$$

$$m^2 - m = 0$$

$$m(m-1) = 0$$

$$m = 0, 1$$

$$C.F = \phi_1(y) + \phi_2(y+x)$$

$$PI = \frac{1}{f(D, D')} F(x, y)$$

$$= \frac{1}{D^2 - DD'} \sin x \cos 2y$$

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$$= \frac{1}{2} \frac{1}{D^2 - DD'} \; 2\sin x \cos 2y.$$

$$= \frac{1}{2} \frac{1}{D^2 - DD'} \; \sin(x+2y) + \sin(x-2y)$$

i) $P.I_1 = \frac{1}{2} \frac{1}{D^2 - DD'} \sin(x+2y)$

$$P.I_1 = \frac{1}{2} \frac{1}{-(1)^2 - DD'} \sin(x+2y)$$

$$= \frac{1}{2} \frac{1}{-(1)^2 - (-2)} \sin(x+2y)$$

$$= \frac{1}{2} \frac{1}{1} \sin(x+2y)$$

$$= \frac{1}{2} \sin(x+2y)$$

ii) $P.I_2 = \frac{1}{2} \frac{1}{D^2 - DD'} \sin(x-2y)$

$$= \frac{1}{2} \frac{1}{-1-(2)} = \frac{1}{-6} \sin(x-2y)$$

$$P.I = \frac{1}{2}\left[\sin(x+2y) - \frac{1}{3}\sin(x-2y)\right]$$

The general solution is $z = C.F + P.I$

$$z = \phi_1(y) + \phi_2(y+x) + \frac{1}{2}\sin(x+2y) - \frac{1}{6}\sin(x-2y)$$

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12. $(D^3 + D^2D' - DD'^2 - D'^3)z = 3\sin(x+y)$

Sol) $(D^3 + D^2D' - DD'^2 - D'^3)z = 3\sin(x+y)$

$f(D, D')z = F(x, y)$

The auxiliary eqn is $f(m, 1) = 0$

$m^3 + m^2(1) - m(1)^2 - (1)^3 = 0$

$m^3 + m^2 - m - 1 = 0$

1	1	1	-1	-1
	0	1	2	1
	1	2	1	0

$(m-1)(m^2 + 2m + 1) = 0$

$(m-1)(m^2 + m + m + 1) = 0$

$(m-1)(m(m+1) + (m+1)) = 0$

$m = 1, -1, -1$

$C.F = \phi_1(y-x) + \phi_2(y-x) + \phi_3(y+x)$

$P.I = \frac{1}{f(D, D')} F(x, y)$

$= \frac{1}{D^3 + D^2D' - DD'^2 - D'^3} 3\sin(x+y)$

$= 3 \frac{1}{D(-1) + (-1)D' - D(-1) - (-1)D'} \sin(x+y)$

$= 3 \frac{1}{-D - D' + D + D'}$

$$= 3x \frac{1}{3D^2 + 2DD' - D'^2} \sin(x+y)$$

$$= 3x \frac{1}{3(-1) + 2(-1) - (-1)} \sin(x+y)$$

$$= 3x \frac{1}{-3 + 2 + 1} \sin(x+y)$$

$$= -\frac{3}{4}x \sin(x+y)$$

The general solution is $z = CF + PI$

$$\Rightarrow z = \phi_1(y-x) + \phi_2 x(y-x) + \phi_3(y+x) - \frac{3}{4}x\sin(x+y)$$

5. $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 6\frac{\partial^2 z}{\partial y^2} = x+y.$

(a) $\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial x \partial y} - 6\frac{\partial^2 z}{\partial y^2} = x+y$

$$(D^2 + DD' - 6D'^2)z = x+y.$$

$$f(D,D')z = x+y.$$

The auxilary eqn is $f(m,1) = 0$

$$m^2 + m(1) - 6(1)^2 = 0$$

$$m^2 + m - 6 = 0$$

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$m^2 + 3m - 2m + 6 = 0$

$m(m+3) - 2(m+3) = 0$

$m = 2, -3$

$C.F = \phi_1(y-3x) + \phi_2(y+2x)$

$$P.I = \frac{1}{f(D,D')} F(x,y)$$

$$= \frac{1}{D^2 + DD' - 6D'^2} \; x+y.$$

$$= \frac{1}{D^2\left[1 + \left[\frac{DD' - 6D'^2}{D^2}\right]\right]} \; x+y$$

$$= \frac{1}{D^2}\left[1 + \left[\frac{DD' - 6D'^2}{D^2}\right]\right]^{-1} x+y$$

$$= \frac{1}{D^2}\left[1 - \left[\frac{DD' - 6D'^2}{D^2}\right]\right](x+y)$$

$$= \frac{1}{D^2}\left[1 - \frac{D'}{D} + \frac{6D'^2}{D^2}\right](x+y)$$

$$= \frac{1}{D^2}\left[x+y - \frac{1}{D} D'(x+y)\right]$$

$$= \frac{1}{D^2}\left[x+y - \frac{1}{D}(1)\right]$$

$$= \frac{1}{D^2}[x + y - x]$$

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$= \frac{1}{D^2}(y)$

$= \iint y \, dx$

$= y\frac{x^2}{2}$

$= \frac{x^2 y}{2}$

The general solution is $z = CF + PI$

$\Rightarrow z = \phi_1(y-3x) + \phi_2(y+2x) + \frac{x^2 y}{2}$

14. $\frac{\partial^3 z}{\partial x^3} - 2\frac{\partial^3 z}{\partial x^2 \partial y} = 2e^{2x} + 3x^2 y.$

Sol) $D^3 - 2D^2D' = 2e^{2x} + 3x^2 y.$

$f(D, D')z = F(x, y)$

The auxilary eqn is $f(m, 1) = 0$

$m^3 - 2m^2(1) = 0$

$m^3 - 2m^2 = 0$

$m(m^2 - 2m) = 0$

$m^2(m-2) = 0$

$m = 2, 0, 0$

$C.F = \phi_1 y + \phi_2 xy + \phi_3(xy + 2x)$

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$$PI = \frac{1}{f(D,D')} F(x,y)$$

$$= \frac{1}{D^3 - 2D^2D'} 2e^{2x} + 3x^2y.$$

$$\Rightarrow PI_1 = 2\frac{1}{D^3 - 2D^2D'} e^{2x}$$

$$= 2\frac{1}{8-0} e^{2x}$$

$$= \frac{1}{4} e^{2x}$$

$$\Rightarrow PI_2 = 3\frac{1}{D^3 - 2D^2D'} x^2y$$

$$= 3\frac{1}{D^3\left[1 - \left[\frac{2D^2D'}{D^3}\right]\right]} x^2y$$

$$= \frac{3}{D^3}\left[1 - \left(\frac{2D'}{D}\right)\right]^{-1} x^2y$$

$$= \frac{3}{D^3}\left[1 + \left(\frac{2D'}{D}\right) + \left[\frac{2D'}{D}\right]^2\right] x^2y.$$

$$= \frac{3}{D^3}\left[1 + \frac{2D'}{D} + \frac{2(D')^2}{D}\right] x^2y$$

$$= \frac{3}{D^3}\left[1 + \frac{2D'}{D}\right] x^2y.$$

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$$= \frac{3}{D^3}\left[x^2y + \frac{2}{D}D'(x^2y)\right]$$

$$= \frac{3}{D^3}\left[x^2y + 2\int x^2\,dx\right]$$

$$= \frac{3}{D^3}\left[x^2y + 2\frac{x^3}{3}\right]$$

$$= 3\iiint\left(x^2y + \frac{2}{3}x^3\right)dx$$

$$= 3\iint\left(\frac{x^3}{3}y + \frac{2}{12}x^4\right)dx$$

$$= 3\int\left(\frac{x^4}{4}y + \frac{2}{60}x^5\right)dx$$

$$= 3\left[\frac{x^5}{5}y + \frac{2}{360}x^6\right]dx$$

$$= 3\left[\frac{x^5}{5}y + \frac{1}{180}x^6\right]dx$$

$$PI = \frac{1}{4}e^{2x} + \frac{3}{5}x^5y + \frac{1}{60}x^6$$

The general solution is $z = CF + PI$

$$\Rightarrow z = \phi_1 y + x\phi_2 y + \phi_3(x+2x) + \frac{1}{4}e^{2x} + \frac{3}{5}x^5y + \frac{1}{60}x^6$$

5. $(D^2 - DD' - 2D'^2)z = (y-1)e^x$

(a) $(D^2 - DD' - 2D'^2)z = (y-1)e^x$

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$f(D,D')z = F(x,y)$

The auxilary eqn is $f(m,1) = 0$

$$m^2 - m(1) - 2(1)^2 = 0$$

$$m^2 - m - 2 = 0$$

$$m^2 - 2m + m - 2 = 0$$

$$m(m-2) + 1(m-2) = 0$$

$$m = -1, 2$$

$$C.F = \phi_1(y-x) + \phi_2(y+2x)$$

$$P.I = \frac{1}{D^2 - DD' - 2D'^2} e^{2}(y-1)$$

$$= e^x \frac{1}{(D+1)^2 - (D+1)D' - 2D'^2} (y-1)$$

$$= e^x \frac{1}{D^2 + 2D + 1 - DD' - D' - 2D'^2} (y-1)$$

$$= e^x \frac{1}{D^2 + 2D - DD' - D' - 2D'^2 + 1} (y-1)$$

$$= e^x \frac{1}{1 + [D^2 + 2D - DD' - D' - 2D'^2]} (y-1)$$

$$= e^x [1 + (D^2 + 2D - DD' - D' - 2D'^2)]^{-1} (y-1)$$

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$= e^x [1 - (D^2 + 2D - DD' - D' - 2D'^2)] (y-1)$

$= e^x [1 - D^2 - 2D + DD' + D' + 2D'^2] (y-1)$

$= e^x [(y-1) + D'(y-1)] = e^x (y - 1 + 1)$

$= e^x (y) \quad e^x y.$

The general solution is $z = CF + PI$

$z = \phi_1(y-x) + \phi_2(y+2x) + e^x y.$

$(D^2 + 2DD' - 8D'^2) z = \sqrt{2x+3y}$

$(D^2 + 2DD' - 8D'^2) z = \sqrt{2x+3y}$

$f(D, D') z = f(x, y)$

The auxiliary eqn is $f(m, 1) = 0$

$m^2 + 2m - 8 = 0$

$m^2 + 4m - 2m - 8 = 0$

$m(m+4) - 2(m+4) = 0$

$m = 2, -2$

$C.F = \phi_1(y-2x) + \phi_2(y+2x)$

$PI = \frac{1}{D^2 + 2DD' - 8D'^2} \sqrt{2x+3y}$

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let $u = 2x+3y$

$a = 2, \quad b = 3$

$$= \frac{1}{4+2(6)-3(9)} \iint \sqrt{u}\, du\, du$$

$$= \frac{1}{4+12-27} \iint (u)^{1/2}\, du\, du$$

$$= -\frac{1}{56} \int \frac{2}{3} (u)^{3/2+1}\, du$$

$$= -\frac{1}{56} \times \frac{2}{3} \left[\frac{1}{5} (u)^{5/2} \right]$$

$$= -\frac{4}{840} (u)^{5/2}$$

$$= -\frac{1}{210} (2x+3y)^{5/2}$$

The general solution is $z = CF + PI$

$$\Rightarrow z = \phi_1(y-2x) + \phi_2(y+2x) - \frac{1}{210}(2x+3y)^{5/2}$$

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